

CSC734: Information Retrieval and Web Search

Fall 2026, Homework Assignment 01

Date Assigned	Friday, September 04, 2026
Due Date	Monday, September 14, 2026 at 11:59 pm
Submission	Brightspace
Total points	120

Objectives:

- Text processing.
- Build the inverted index using a collection of documents.

Tasks (100 points)

For this assignment, you need to create an environment with the specified requirements. Download the dataset documents.zip from Brightspace. Write the Python code that:

1. Read and process text dataset.
 - Read the text.
 - Tokenize.
 - Remove stop words (you must use the provided list)
 - Normalize (lower case, remove punctuation, stem, ...)
2. Write your own code to build the inverted index.
3. Displays the size of the index (in byte and MB).
4. Display the top n frequent terms (in the entire collection).
5. Save the index in a file/load the saved index file.

Important (20 points)

- Your code should have at least five functions. One function should display course and student information as follows:

```
===== CSC734-IR Homework 01 =====  
First Name: your first name  
Last Name : your last name  
=====
```
- Document your code: write comments to explain the role each function and block of code; what are the input parameters and the output/return parameters.
- The code must be well organized and easy to use.

- You must use NLTK.

Submission

1. Write your own code. Use as many functions as you can.
2. Make sure you writing you name and assignment number on all files you submit.
3. Your python code and the instructions on how to run it.
4. Enclose all your files in a folder named **HW01__yourlastname.zip**.
5. Submit the zip file using Brightspace.